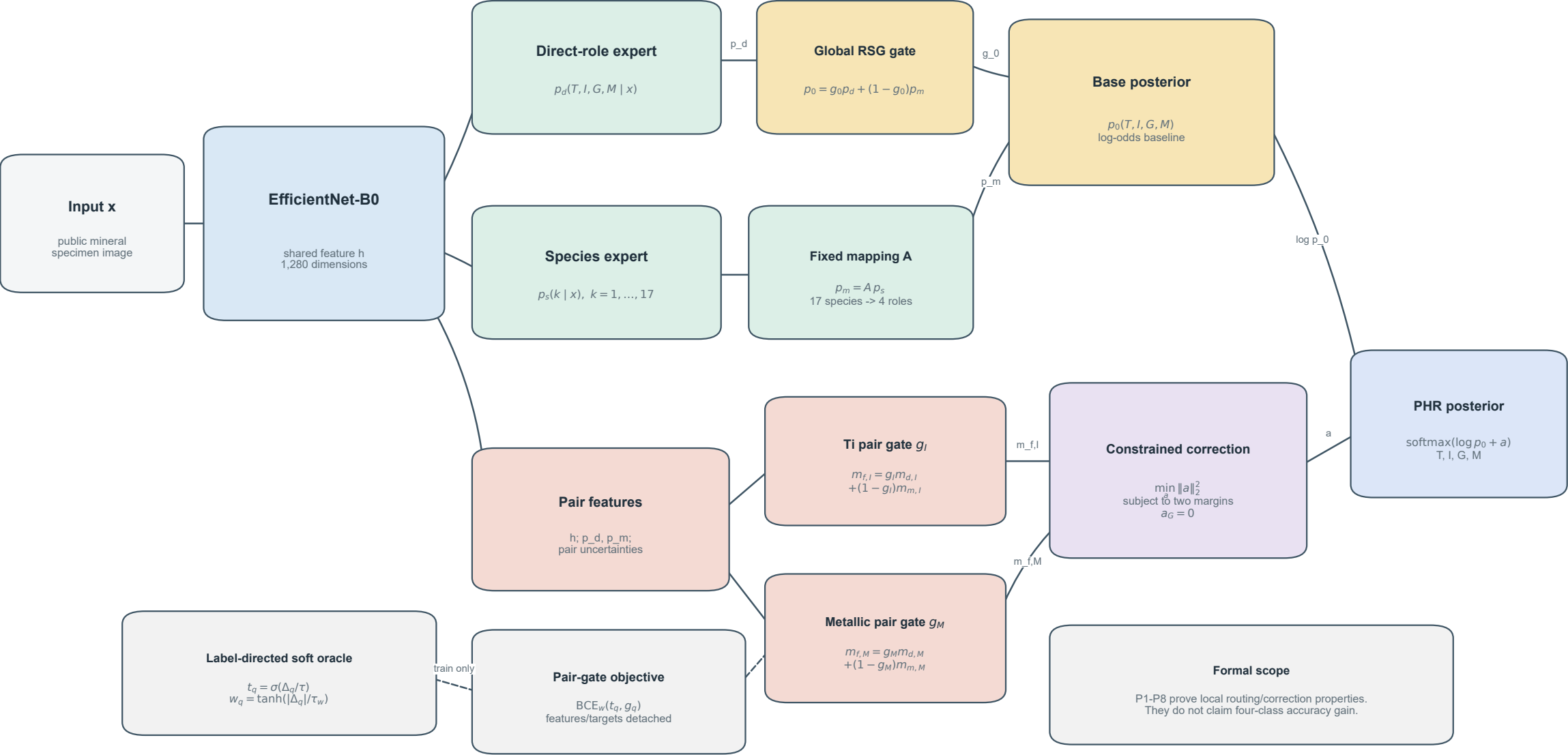


PHR-Routing-Net: pairwise hard-negative regret routing (candidate extension)

Two target-versus-negative margins are routed independently; dashed paths and oracle labels are training-only.



Roles: T target proxy; I Ti-bearing negative; G gangue; M metallic hard negative. Candidate architecture: no empirical performance claim.